B.Tech Curriculum

B.Tech (2026) B.Tech(2025) B.Tech (2024) B.Tech (2023) Elective (B.Tech) Minor in AI (B.Tech)

Curriculum for B.Tech in Artificial Intelligence (2025 onwards)

Year	Odd Semester		Credits	Even Semester		Credits
1	MA1110	Calculus I	1	MA1150	Differential Equations	1
	MA1220	Calculus II	1	MA1230	Series of Functions	1
		Science Basket (See below)	2	AI1020	Intro to AI	3
	ID1063	Programming	3	AI1110	Probability and Random Variables	3
	AI1000	Matrix Theory	3	AI1233	Optimization - I	3
	CS1010	Discrete Math for Computer Science	3	AI1013	Programming for AI	2
	BT1010	Introduction to Life Sciences	1		LA/CA Elective	2
	LA1760	Communication Skills	2			
	Total		**16**	**Total**		**15**
2	ID2230	Data Structures & Applications	3	CS2443	Algorithms	3
	MA2150	Introduction to Metric Spaces	1	AI2100	Deep Learning	3
	MA3060	Numerical Analysis	3	EE2101	Control Systems	3
	AI2000	Foundations of ML	3	AI2113	Optimization - II	3
	CS3550	DBMS - I	1		LA/CA Electives	1
		Science Basket (See below)	2	AI3013	AI for Humanity	3
	EE1206	Linear Systems and Signal Processing	3			
	AI2200	Concentration Inequalities	1			
	Total		**17**	**Total**		**16**
3	AI3000	Reinforcement Learning	3		Free Electives	3
	AI4000	Robotics	3	AI3703	Natural Language Processing	3
	AI3005	AI Project	3	AI3603	Computer Vision	3
		LA/CA Electives	1	MA2570	Applied Statistics	3
		AI Elective (See Baskets below)	3		LA/CA Elective	1
		Free Electives	3		AI Elective	3
	Total		**16**	**Total**		**16**
4		AI Electives (see baskets)	9		AI Electives (see baskets)	15
		Internship*		LAxxxx	Ethics and Values	1
		LA/CA Elective	1			
		Free Elective	6			
	Total		**16**	**Total**		**16**

*(6 credits can be credited from internship)

1. Out of 27 department electives, 15 must be from the baskets (as specified below). The remaining 12 credits can be any of the remaining basket courses.
2. Six credits of Department Electives in the seventh semester can optionally be converted to a semester-long internship in the seventh semester. The onus is on the student to distribute/complete the remaining 9 credits in the eighth semester.
3. Electives not in the baskets below can be considered in a given basket with the approval of faculty advisor (e.g. a new AI elective offered by a new faculty).
4. Mini-project courses AI 3015, AI 4005, AI 4015 can be taken as followup to AI3005 in Sem 6,7,8 respectively as department electives.

Category	Credits	Percentage
Free Elective	12	9.38%
Basic Science	16	12.50%
Basic Engineering Skills	16	12.50%
Department Elective	30	23.44%
Department Core	45	35.16%
LA/CA	9	7.03%
Total	**128**	**100%**

Science Basket

Course Code	Course Name
EP1108	Modern Physics
CY1010	Environmental Chemistry
MA2130	Complex Variables
MA2120	Transform Techniques
MA2150	Introduction to Metric Spaces*
EP2200	Thermodynamics*
EP2108	Special Relativity*
EP2228	Fluid Dynamics*
EP2100	Classical Mechanics*
CC1010	Introduction to Climate change*
MA1240	Combinatorics*
MA2140	Introduction to Statistics*
MA2510	Linear Programming*

*Subject to approval of parent department

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